

MAJOR CONTRIBUTORS TO THE STORY



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Ambrose Popper, VP, Mobile Computing and M2M business unit, Sequans Communications, says, "Although it only represents a small portion of the overall M2M market, demand for LTE-enabled M2M is growing fast today. By the year 2017, one market analyst predicts, 70 per cent of M2M modules will be LTE." He adds, "Driving M2M applications for LTE are metering, home-automation, automotive accessories and wearables."

Although carriers and LTE module makers are focused on the M2M market for looking beyond people-centric mobile communications, the cost of deployment and operation is a deterrent, feels Venkatesh. He says, "Greater demand is for modules that offer wireless communication in unlicensed spectrum—protocols applicable being Wi-Fi, Bluetooth 4.0, ZigBee, ANT+, to name a few." He adds, "It remains to be seen what the uptake on LTE-U (LTE in the unlicensed 5GHz band) would be."

Talking about implementation of LTE for IoT solutions in the current Indian scenario, Darshan Virupaksha, systems engineer, Altix Innovations & co-organiser IoTBLR, feels, it is a milestone set too far to achieve. He says, "Although the all-IP architecture of LTE is a perfect fit for the IoT, the ecosystem is not ready yet. The upcoming government spectrum auctions might change the scenario." He adds, "A couple of module vendors have 4G modems in the market, and one can say that, the demand has not attained critical mass. Telematics solutions and large-scale implementa-

Modules v/s chipsets: Summary

Modules are relatively more expensive and have a faster TTM. At risk is scalability and lack of supplier redundancy. Lesser development budgets and resources automatically point to modules.

Chipsets are cheaper, but development effort and TTM is more. Can be a distinct advantage in a cost-sensitive volume segment. Inter-changeability and second supplier redundancy gains merit here.

— **Arati Mekaraj**, deputy general manager (DGM) – IOT, Robert Bosch Engineering and Business Solutions

tions like smart cities could be one of the first adopters of 4G in India."

M2M chipset or an M2M module—how to decide

For low volumes, like less than 50,000 units, modules would be best-suited, feels Virupaksha. On the other hand, he says, "Chipsets come with low per-unit cost, and there would be high certification costs included. It would be viable to go for chipsets only for volumes around one million or so."

Certification is a key factor to be considered by designers for today's chipsets and modules. M2M modules come pre-certified. Praveen Ganapathy, director, business development, TI India says, "It could take four to six months for a designer to get his design certified if he starts from scratch. Having a certified module gives a greater edge, especially for a start-up looking to do a quick proof-of-concept, evaluate and get it to the market quickly." He adds, "It really is a balance between volume, TTM and cost."

Some other benefits associated with using modules are shorter development period, higher per-unit

costs, ruggedness and reliability. A bare minimum of RF expertise is required to work with modules. Popper says, "A module requires much less research and development expertise from a developer when compared to a chipset." He adds, "For carriers with hard certification programs, a pre-certified module makes more sense."

What is the biggest reason people would go for modules, questions Singh. He explains that, in the software world, it is very common to use a very high-optimised module and library if something is going to be repeated over and over again. This is transitioning to the hardware world as well. Singh adds, "A module that has been built for BLE, ZigBee or Wi-Fi is packed immensely dense, such that the size and power consumption goes down significantly. If I am building my solution from scratch, I would probably not be able to get that level of integration and power-consumption. So, that is why I prefer using modules."

Time-to-market is a very important consideration to make, informs Ganapathy. He says, "If you have a